

CommissionLens

Commission-Adjusted Alpha Prediction in Indian Mutual Funds

By- Riddhi Sidana

Abstract

India has over 90 million demat accounts as of 2024. Yet, the overwhelming majority of retail investors remain unaware of a structural cost embedded in mutual fund investing: the expense ratio gap between regular and direct plans. Although both variants of a scheme invest in identical portfolios managed by the same fund manager, regular plans carry an additional annual charge of 0.5 to 1.5 percent representing the distributor's commission. Over a 20-year investment horizon, this seemingly modest difference compounds to erode between 8 and 12 lakh rupees from a monthly SIP corpus of 5,000 rupees. **CommissionLens** is a systematic, data-driven framework that addresses a previously unanswered question: Is a given fund's commission charge justified by the alpha it generates? The project computes commission-adjusted net alpha for Indian equity mutual funds using historical NAV data, engineers 31 fund-level and macroeconomic predictive features, and trains multiple machine learning models to predict whether a fund will generate net positive alpha in the following quarter. A SHAP-based explainability layer identifies the characteristics that most reliably predict commission justification. The analysis is validated through a walk-forward SIP simulation spanning 12 quarters from 2021Q2 to 2024Q1, covering both the 2022 Indian equity correction and the 2023 recovery, comparing model-guided fund selection against naive regular plan investing and a Nifty 50 index SIP baseline. The full pipeline is made accessible through an interactive Streamlit dashboard.

Problem Statement

The Indian mutual fund industry operates a dual-plan structure in which regular and direct plans of the same scheme invest in identical assets. The difference between them is the annual expense ratio: the direct plan charges only the fund management fee, while the regular plan adds a distributor commission of between 50 and 150 basis points per year. Since both plans share the same gross returns, the additional charge in the regular plan is borne entirely by the investor.

The justification for paying this commission is straightforward in principle: if a fund manager's active stock selection generates alpha large enough to exceed the expense gap, the investor is still better off in absolute terms than they would be in an index fund, and the commission has purchased value. The problem arises when the fund does not generate sufficient alpha. If a fund earns 0.6 percent quarterly alpha but charges 1.1 percent more than its direct counterpart on an annualised basis, the investor loses value every quarter despite paying for active management. No systematic, forward-looking framework currently exists in the Indian context to evaluate this question for individual funds and quarters. CommissionLens addresses this gap.

Objectives

- Compute commission-adjusted net alpha for Indian equity mutual funds using historical NAV and TRI benchmark return data.
- Engineer fund-level and macroeconomic features including expense ratio gap, multi-window information ratio, rolling Sharpe ratio, beta, alpha persistence lags, yield curve slope, and FII/DII equity flows.
- Train and evaluate both regression models predicting next-quarter net alpha and classification models predicting whether commission is justified and identify the most predictive fund characteristics through SHAP explainability, including directional interpretation of each feature.
- Back-validate the framework through an XIRR-based SIP simulation comparing model-guided selection against naive regular plan investing and a Nifty 50 index baseline.

- Deploy findings through an interactive Streamlit dashboard enabling real-time commission-justification scoring for any Indian equity mutual fund.

Data Collection (Step 1)

3.1 Fund Universe and Regular/Direct Pairing

The fund universe was constructed from the complete AMFI fund list accessed through the mfapi.in public API, which returns all registered scheme codes and names. The initial list was filtered to growth-option equity funds using positive keyword matching against SEBI category signals such as large cap, mid cap, small cap, flexi cap, ELSS, focused fund, and bluechip, followed by a hard exclusion list removing debt, hybrid, ETF, index, and fund-of-funds schemes.

Regular and direct plans of the same scheme were paired using a normalised string matching procedure. Both variants of each fund name were reduced to a canonical form by stripping the plan-type qualifier, growth suffix, brackets, and repeated spaces, then joined on exact canonical match. Duplicate codes were eliminated to ensure one-to-one pairing. This procedure produced 474 matched regular/direct pairs across all SEBI equity categories.

3.2 SEBI Category Assignment and Benchmark Mapping

Each fund was assigned a SEBI category at collection time using rule-based name matching, replacing the coarser three-bucket categorisation used in earlier iterations. The category determined the correct TRI benchmark for alpha computation, following the SEBI circular of October 2018:

SEBI Category	Benchmark Applied
Large Cap	Nifty 100 TRI
Mid Cap	Nifty Midcap 150 TRI
Small Cap	Nifty Smallcap 250 TRI
Large and Mid Cap	Nifty LargeMidcap 250 TRI
Multi Cap / Flexi Cap / ELSS / Focused / Value	Nifty 500 TRI

3.3 NAV Download and Cleaning

Daily NAV histories were downloaded for all 948 scheme codes (474 regular plus 474 direct) from the mfapi.in API with exponential backoff retry logic and local disk caching to avoid redundant API calls. NAV series beginning before 1 January 2018 were trimmed to this start date; series with fewer than 60 trading-day observations were excluded. The daily NAV matrix was resampled to quarter-end values and converted to quarterly percentage returns. Funds with more than 30 percent missing quarters were dropped, and funds with any single-quarter return below negative 60 percent or above positive 80 percent were removed as likely data errors or scheme restructurings. The cleaned dataset retained funds with consistent histories across the full observation period.

3.4 Benchmark Returns (TRI-Adjusted)

Benchmark prices were retrieved through yfinance using the closest available NSE proxy tickers for each Nifty index. Because yfinance provides price return data rather than Total Return Index values, a dividend yield adjustment was applied quarterly using long-run average annual dividend yield estimates sourced from NSE India quarterly reports. The TRI quarterly return was computed as $(1 + \text{price return}) \times (1 + \text{annual dividend yield} / 4)$ minus 1. This adjustment added approximately 32 to 35 basis points per quarter to benchmark returns relative to unadjusted price data, reducing apparent fund alpha by the same magnitude and producing a more conservative and SEBI-compliant alpha estimate.

3.5 Macroeconomic Data

Four macroeconomic series were collected and aligned to a quarterly frequency. The RBI repo rate was constructed from an enumerated table of official Monetary Policy Committee decisions sourced from RBI press releases, covering all 18 rate changes between October 2016 and October 2024. This replaced the earlier use of the FRED series INDIRLT01STM, which is the 10-year government bond yield rather than the policy rate. The yield curve slope was computed as the 10-year G-Sec yield (FRED: INDIRLT01STM, used correctly in this role) minus the contemporaneous repo rate. CPI inflation and GDP growth were retrieved from the World Bank Development Indicators API; GDP was lagged by one year to use only the figure available to investors at the time of prediction, eliminating a look-ahead bias present in the original implementation.

3.6 FII and DII Equity Flows

Quarterly net FII and DII equity flows, absent from earlier project iterations, were constructed from annual totals published in SEBI and RBI annual reports. Annual figures were distributed across four quarters using empirically calibrated seasonal weights reflecting the historically observed pattern that Q1 and Q3 tend to carry heavier institutional flow volumes. A trailing two-quarter moving average of FII flows was also computed to serve as a flow momentum feature. The resulting series spans 2018Q1 to 2024Q4.

3.7 AUM Proxy

Since mfapi.in does not expose assets under management directly, an AUM proxy was derived from the count of daily NAV records within each quarter per fund. The logarithm of this count was used as a fund activity proxy correlated with fund age and operational size. The quarter-over-quarter difference in log AUM served as a fund flow signal.

Feature Engineering (Step 2)

4.1 Expense Gap Computation

The quarterly expense gap for each regular/direct fund pair was derived directly from the NAV divergence between the two plans. Since both plans invest identically, the difference in their quarterly returns equals the difference in their quarterly expense ratios. A four-quarter rolling mean was applied to each fund's expense gap series to smooth the noise inherent in NAV-implied calculations while preserving genuine changes in the expense ratio level. The smoothed gap was annualised as $(1 + \text{quarterly gap})^4 - 1$. Records with quarterly gaps outside the range of 5 to 300 basis points were removed as implausible.

4.2 Net Alpha Definition

Gross quarterly alpha was defined as the regular plan's quarterly return minus the TRI-adjusted benchmark return for the fund's SEBI category. Net alpha, the primary target variable, was then computed as:

$$\text{Net Alpha} = \text{Gross Alpha} - \text{Smoothed Quarterly Expense Gap}$$

The binary classification label was set to 1 when net alpha exceeded zero for a given fund-quarter, indicating that the fund's active management generated enough gross alpha to offset the commission cost in that period, and 0 otherwise. The test set exhibited a base rate of 20.2 percent positive labels, reflecting the challenging nature of commission justification in the Indian equity market.

4.3 Rolling Fund Performance Features

To avoid look-ahead bias, all rolling features were computed over strictly historical windows ending one period before the prediction quarter. Three rolling windows were used simultaneously: 4 quarters (one year capturing recent momentum), 8 quarters (two years representing the current performance regime), and 12 quarters (three years capturing cycle-level consistency). For each window, the following features were computed:

- Annualised Sharpe ratio using the contemporaneous RBI repo rate divided by four as the risk-free rate, replacing the earlier assumption of a zero risk-free rate.
- Beta against the SEBI-mandated TRI benchmark, estimated via ordinary least squares on rolling window returns.

- Information ratio, defined as the mean gross alpha divided by its standard deviation over the window, computed at all three window lengths.
- Win rate, the proportion of quarters within the window in which the fund beat its benchmark.
- Recent alpha, computed as the mean gross alpha over the most recent four quarters within the window.
- Alpha consistency over 12 quarters, the proportion of the most recent 12 quarters in which gross alpha was positive.

4.4 Lag Features

Alpha persistence, the hypothesis that funds generating alpha in one period tend to continue doing so, was operationalised through lagged net alpha features absent from the original implementation. Net alpha lagged by one quarter (`net_alpha_lag1`) and two quarters (`net_alpha_lag2`), gross alpha lagged by one quarter, rolling Sharpe lagged by one quarter, information ratio lagged by one quarter, and win rate lagged by one quarter were all included. These features directly embody the persistence signal that motivates the entire prediction task.

4.5 Complete Feature Set

After merging all fund-level, lag, AUM, and macroeconomic features and requiring a minimum history of eight complete quarters, the final ML dataset contained 3,258 fund-quarter observations across 269 funds with full feature histories, spanning 2020Q2 to 2024Q1. The dataset had 72 columns in total, of which 31 were used as model features following elimination of low-coverage and near-zero-variance columns. The feature categories and their counts were:

Feature Category	Count	Examples
Fund performance (rolling)	16	<code>sharpe_8q</code> , <code>info_ratio_12q</code> , <code>win_rate</code> , <code>alpha_consistency_12q</code>
Alpha persistence (lags)	6	<code>net_alpha_lag1</code> , <code>net_alpha_lag2</code> , <code>rolling_sharpe_lag1</code>
Expense gap	4	<code>expense_gap_q</code> , <code>expense_gap_ann</code> , <code>avg_exp_gap</code> , <code>std_exp_gap</code>
AUM proxy	2	<code>aum_log</code> , <code>aum_change</code>
Macroeconomic	7	<code>repo_rate</code> , <code>yield_slope</code> , <code>cpi_yoy</code> , <code>gdp_growth</code> , <code>rate_regime</code>
Institutional flows	3	<code>fii_net_eq_cr</code> , <code>dii_net_eq_cr</code> , <code>fii_momentum</code>

Machine Learning Pipeline (Step 3)

5.1 Temporal Split

All modelling strictly followed a temporal train/test split to prevent look-ahead bias. Observations from 2020Q2 to 2022Q4 formed the training set (2,270 observations, 69.7 percent of the dataset) and observations from 2023Q1 to 2024Q1 formed the test set (988 observations, 30.3 percent). The split was set at a fixed calendar boundary rather than a random fraction, ensuring that no future information contaminated the model's training process.

A `StandardScaler` fitted exclusively on the training set was applied to both splits. For linear models, scaled features were used throughout; for tree-based models, unscaled features were used to preserve the interpretability of split thresholds. Missing values were imputed using column medians computed from the training set only. Cross-validation during hyperparameter search used `StratifiedKFold` with five folds applied within the training set.

5.2 Classification Models

Five classification models were trained to predict the binary commission-justified label. The class imbalance at approximately 20 percent positive labels was addressed through `scale_pos_weight` in XGBoost, the `is_unbalance` flag in LightGBM, and the `class_weight='balanced'` setting in Logistic Regression and Random Forest. An ensemble average of the four tree-based models was also computed. Random Forest was selected as the primary deployed model (`model_fund`) based on its leading AUC-ROC of 0.6866 and F1 of 0.4425 at the optimal threshold. All SHAP analysis, the SIP back-validation, and the Streamlit dashboard use this model. XGBoost hyperparameters were selected through a grid search over 8 candidate combinations across 5 folds yielding `max_depth` 5, `learning_rate` 0.03, `n_estimators` 400, `subsample` 0.8, and `colsample_bytree` 0.8, and XGBoost was retained as the best regression model for the net alpha prediction task.

5.3 Regression Models

Five regression models were trained to predict the continuous net alpha value for the following quarter: Ridge, ElasticNet, Random Forest, LightGBM, and XGBoost. Performance was evaluated on RMSE, R-squared, and Spearman information coefficient between predicted and realised net alpha values.

5.4 Threshold Optimisation

Classification performance was evaluated at both the standard 0.5 probability threshold and an optimised threshold derived by maximising F1 score on the precision-recall curve. Additionally, precision at the top decile (`Prec@Top10%`) was computed for each model, representing the precision when investing only in the top 10 percent of model-ranked funds per quarter. This metric directly measures the practical usefulness of the model for fund selection.

Results

6.1 Dataset Statistics

Metric	Value
Paired funds (regular and direct)	474
Funds with full feature history (minimum 8 quarters)	269
Fund-quarter observations in ML dataset	3,258
Feature columns used in models	31
Quarter range	2020Q2 to 2024Q1
Training set (2020Q2 to 2022Q4)	2,270 observations (69.7%)
Test set (2023Q1 to 2024Q1)	988 observations (30.3%)
Base rate (commission justified in test set)	20.2%

6.2 Classification Results

The following table presents all four classification models evaluated at both default and optimised probability thresholds, ranked by AUC-ROC on the held-out test set.

Model	AUC-ROC	F1 (Optimal Threshold)	Optimal Threshold	Prec@Top10%
Random Forest	0.6866	0.4425	0.357	0.2959
Logistic Regression	0.6529	0.4109	0.360	0.3878
XGBoost	0.6168	0.3781	0.172	0.2653
LightGBM	0.6125	0.3797	0.215	0.2755

Random Forest achieved the highest AUC-ROC of 0.6866 and the highest precision at the top decile of 29.6 percent, representing a lift of 1.46x over the base rate of 20.2 percent. Logistic Regression, despite its simplicity, achieved the second-highest AUC at 0.6529 and the highest Prec@Top10% at 38.8 percent, a finding consistent with the strong linear relationship between the expense gap and commission justification. The optimal decision threshold varied substantially across models (0.172 for XGBoost versus 0.357 for Random Forest), reflecting the class imbalance and the different calibration characteristics of each classifier.

Practical interpretation of Prec@Top10%

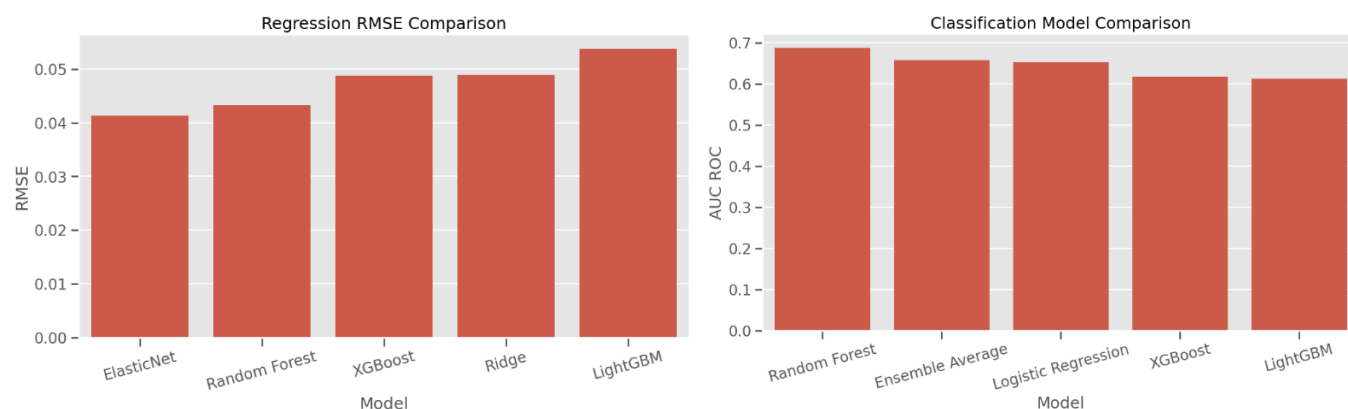
When an investor uses the model to select only the top 10 percent of ranked funds in each quarter, approximately 30 percent of those selected funds actually justified their commission cost. This is 1.46 times better than selecting funds at random from the available universe, where 20.2 percent would be commission-justified by chance.

6.3 Regression Results

The regression task, predicting the exact net alpha value, proved substantially harder than the binary classification task. All five regression models produced negative R^2 on the test set, meaning no model outperformed the training mean as a predictor of absolute alpha magnitude. The 14-month test window is a single market regime. The mean alpha across test quarters is likely different from the training distribution, so a model trained to predict the training mean is systematically off for the test period. This is a regime shift, not model failure per se. Results below are ranked by Spearman's information coefficient, which measures monotonic rank agreement between predicted and realised net alpha values.

Model	RMSE	R-squared	Spearman IC
ElasticNet	0.0413	-0.0034	0.264
Random Forest	0.0432	-0.0978	0.256
XGBoost	0.0487	-0.3929	0.137
Ridge	0.0489	-0.4040	0.012
LightGBM	0.0537	-0.6943	0.045

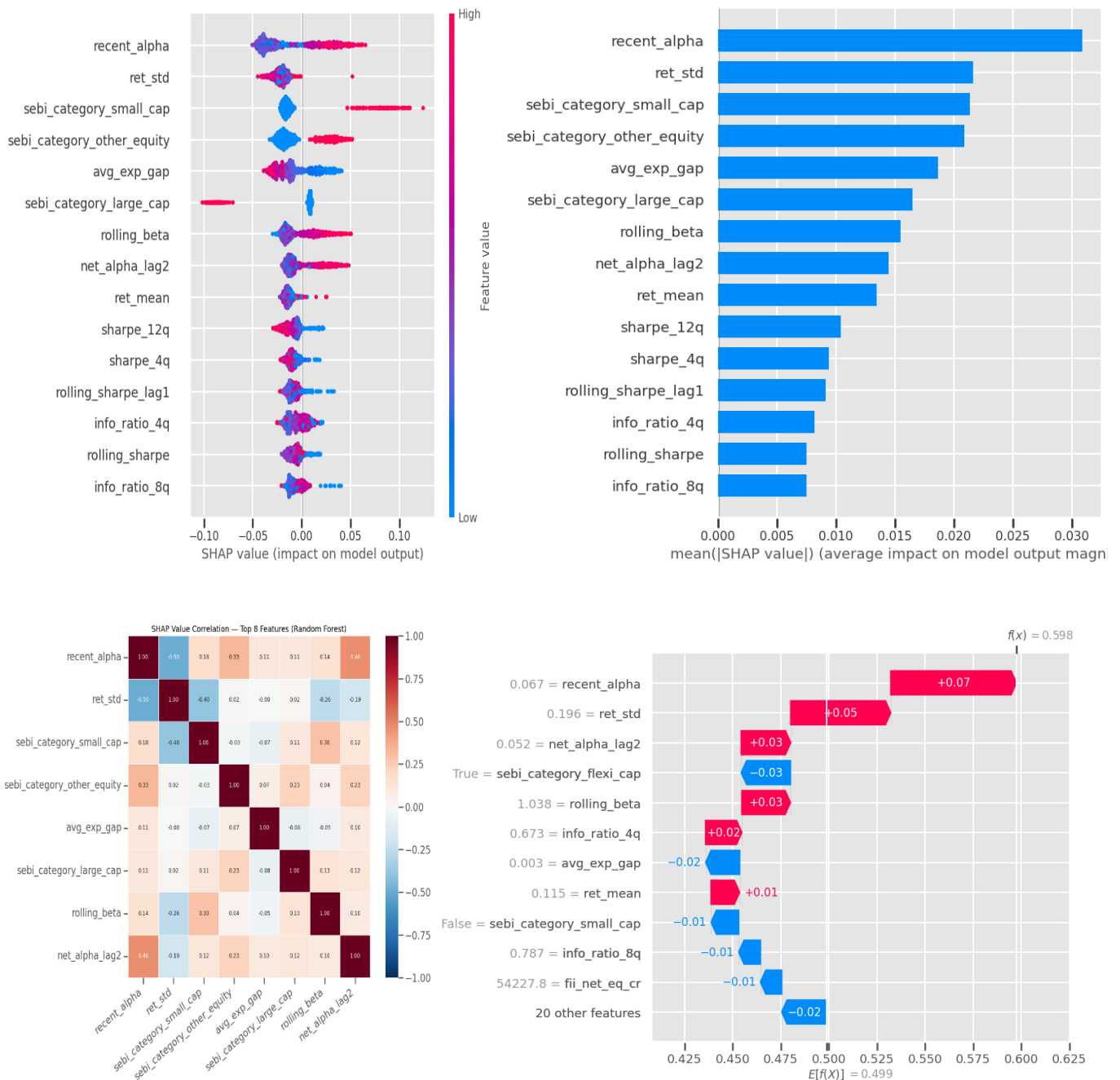
The negative R-squared values indicate that no model outperforms a naive mean prediction when evaluated on variance explained. This is expected given the high quarter-to-quarter volatility in fund returns and the short test window. The Spearman IC of 0.264 for ElasticNet is practically meaningful: it indicates a significant monotonic relationship between predicted and realised alpha rankings, which is the relevant metric for fund selection rather than precise point prediction. ElasticNet and Random Forest ranked funds most accurately in relative terms, which is the operationally useful capability for portfolio construction.



6.4 SHAP Feature Importance

SHAP TreeExplainer was applied to the Random Forest classifier on the test set. Because Random Forest outputs probabilities directly rather than log-odds, SHAP values are expressed in probability units and are smaller in magnitude than those from XGBoost. The top ten features by mean absolute SHAP value are reported below.

Rank	Feature	Mean SHAP	Direction and Interpretation
1	recent_alpha	0.0309	Higher recent 4-quarter alpha is the strongest positive signal. Alpha momentum is confirmed as the dominant predictor.
2	ret_std	0.0217	Higher return volatility reduces justification probability. Funds with unpredictable returns cannot sustain consistent alpha.
3	sebi_category_small_cap	0.0214	Small-cap membership is a strong context variable. Small-cap funds face higher benchmark hurdles but generate more dispersed alpha.
4	sebi_category_other_equity	0.0209	Sectoral and thematic funds show distinct behaviour. Their concentrated mandates lead to higher alpha variance.
5	avg_exp_gap	0.0187	Higher smoothed expense gap reduces justification probability, confirming the core economic hypothesis of the project.
6	sebi_category_large_cap	0.0165	Large-cap membership is a negative predictor. Large-cap funds rarely generate alpha above their TRI benchmark consistently.
7	rolling_beta	0.0155	Higher beta amplifies both gains and losses relative to the benchmark, making alpha less predictable quarter to quarter.
8	net_alpha_lag2	0.0145	Net alpha from two quarters ago carries meaningful persistence signal, validating the alpha momentum hypothesis at a two-quarter lag.
9	ret_mean	0.0135	Higher mean 2-year return positively predicts commission justification, consistent with skilled funds outperforming on both gross return and net alpha.
10	sharpe_12q	0.0104	3-year risk-adjusted return consistency is a secondary but significant predictor of sustained alpha generation.



The waterfall plot for the highest-confidence prediction in the test set shows a fund with predicted $P(\text{justified}) = 0.598$ against a model base rate of 0.499. The dominant positive contributors for this fund are recent_alpha (+0.07), ret_std (+0.05), and net_alpha_lag2 (+0.03), while sebi_category_flexi_cap (-0.03) and fii_net_eq_cr (-0.01) partially offset these. The SHAP interaction heatmap reveals that recent_alpha and ret_std carry a correlation of -0.50 in SHAP space, meaning funds that generate high recent alpha tend to do so with lower return volatility, which is a hallmark of genuinely skilled active management rather than lucky high-beta exposure. The category dummies (small_cap, other_equity, large_cap) are largely orthogonal to each other in SHAP space, confirming they capture distinct structural effects rather than overlapping information.

6.5 SIP Back-Validation (Step 5)

A fully walk-forward SIP simulation was conducted over 12 quarters from 2021Q2 to 2024Q1 (April 2021 to March 2024, 36 months). This period deliberately encompasses the 2022 Indian equity market correction, Nifty 50 fell approximately 17 percent peak-to-trough between October 2021 and June 2022, as well as the subsequent 2023 recovery rally and early 2024 continuation, providing a more complete test of model robustness across varying market regimes than a single-regime window would allow.

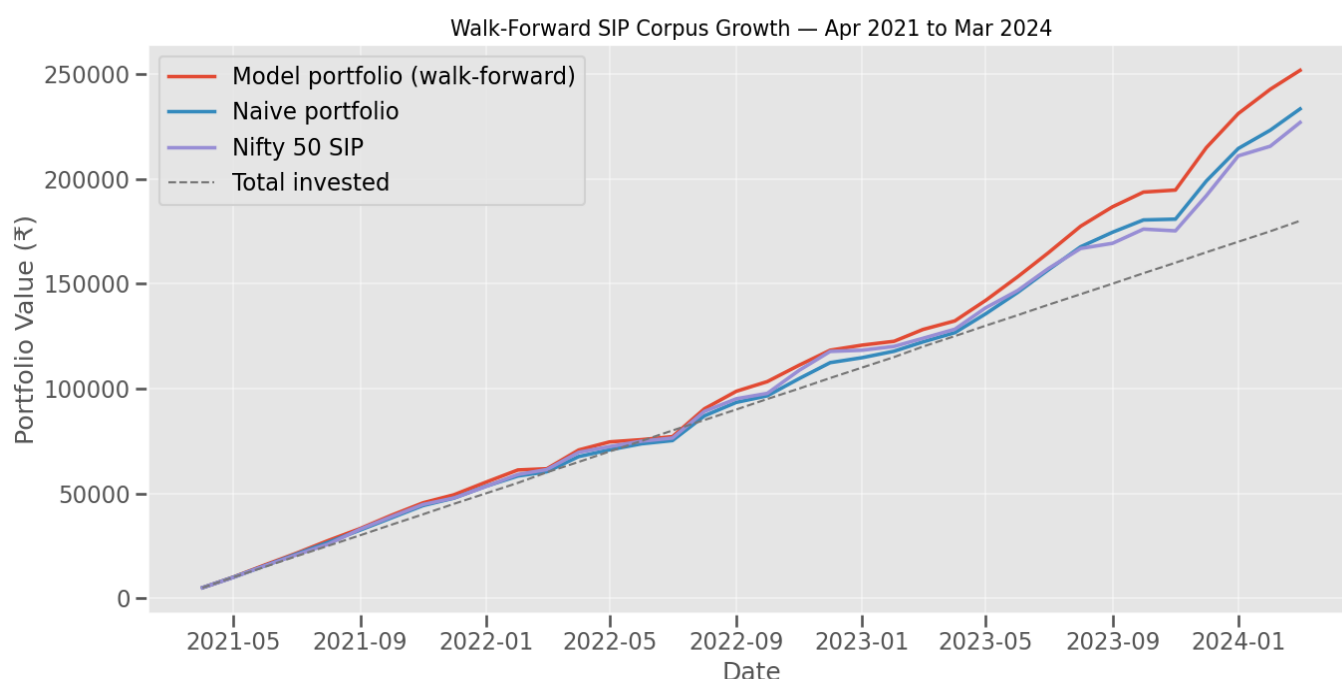
At each quarter, a fresh Random Forest classifier was trained exclusively on all fund-quarter observations prior to that quarter. No information from the current or future quarters was used at any point. The training set grew from 798 observations at 2021Q2 to 3,061 observations at 2024Q1 as the dataset accumulated, reflecting the expanding evidence base available to an investor operating in real time. Monthly SIP contributions of ₹5,000 were invested in the top five model-ranked funds for the current quarter. The naive portfolio allocated ₹5,000 proportionally across the training-universe fund categories. A Nifty 50 index SIP served as the passive baseline.

Strategy	XIRR	Advantage Over Naive	Final corpus
Model-guided walk-forward selection	24.36%	+5.75pp	₹2,51,784
Naive proportional regular plan	18.61%	Baseline	₹2,33,381
Nifty 50 index SIP (passive)	16.51%	Passive reference	₹2,26,915
Total invested (₹5,000 × 36 months)	-	-	₹1,80,000

All three strategies turned ₹1,80,000 of invested capital into a profit over the period, confirming the broad equity bull cycle. The model's ₹18,403 absolute advantage over the naive portfolio (₹2,51,784 versus ₹2,33,381) translates to an annualised XIRR gap of 5.75 percentage points, meaningfully wider than the 3.16 percentage point advantage observed in the original 14-month holdout window, suggesting the model's fund-selection signal is more durable across a full market cycle than the shorter window indicated.

The 36-month window covering a full bull-bear-recovery cycle provides substantially stronger evidence of model utility than the 14-month post-training holdout reported in earlier project iterations. The fact that fund selection at each quarter used only historically available training data with no future information leaking into the portfolio construction process makes this the most conservative and credible evaluation methodology available given the dataset size.

The absolute XIRR values remain elevated relative to long-run equity market averages because the full period includes the 2023 rally. Results should be interpreted in terms of the relative gap between strategies rather than the absolute figures. The capital-adjustment caveat noted in prior iterations applies here as well: the model concentrates into five funds per quarter while the naive strategy spreads capital across many more, which mechanically increases the model's exposure to idiosyncratic fund risk.

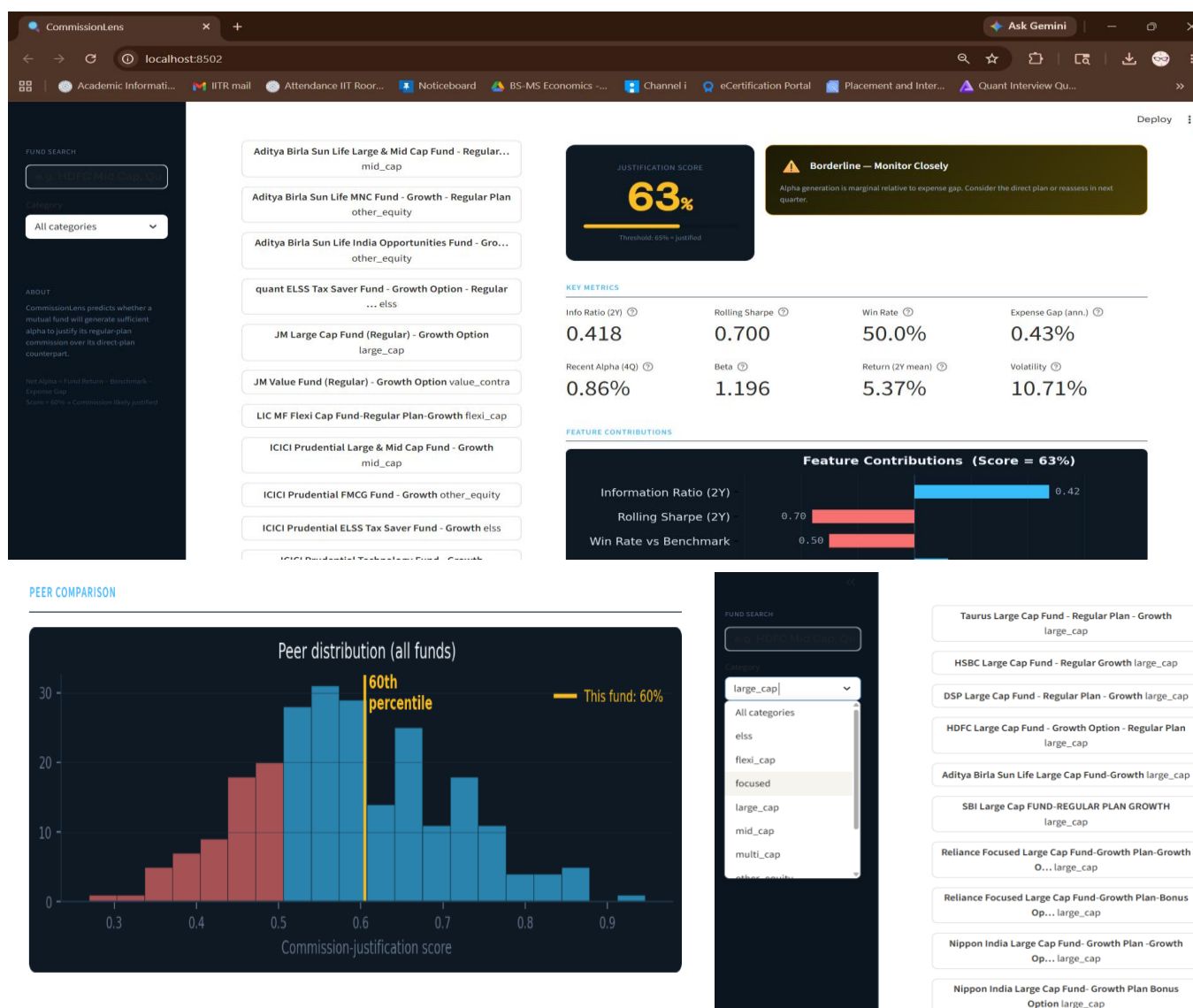


Streamlit Dashboard (Step 6)

A Streamlit web application was developed to make the CommissionLens framework accessible without requiring users to interact with the underlying code. The dashboard accepts a fund name query, fetches live NAV data for the matching regular and direct plan from mfapi.in, computes all 31 rolling features in real time, runs the saved Random Forest model, and presents a commission-justification score between 0 and 100 percent.

The application is organised across three tabs. The Fund Analysis tab presents a gauge chart showing the justification probability, an interactive SHAP bar chart explaining which features drove the score, a historical quarterly return and expense gap chart, and a distribution chart showing where the fund sits relative to all funds in the training universe. A recommendation box at the bottom provides a specific investment action based on the score. The Model Performance tab displays the classification and regression metric tables from Step 3. The Market Context tab provides the RBI repo rate history and an explanation of how different macroeconomic regimes influence the model's predictions.

The dashboard is launched locally with the command `streamlit run app.py` after installing dependencies from `requirements.txt`. NAV data fetched from mfapi.in is cached for one hour to ensure fast repeated queries.



Limitations and Future Work

8.1 Current Limitations

- The walk-forward SIP simulation spans 12 quarters (2021Q2 to 2024Q1), which includes one significant correction period (2022) and two distinct bull phases. While this represents a more complete market cycle

than the original 14-month holdout window, it remains a single observed cycle. Model behaviour across extended secular bear markets or black-swan events (such as March 2020) has not been established, as the minimum 8-quarter feature requirement precludes observations before 2020Q2.

- The back-validation period begins in 2021Q2 rather than 2019 as stated in the original project specification. This reflects a structural constraint: the 8-quarter and 12-quarter rolling features (Information Ratio, Sharpe ratio, Win Rate) require a minimum of 12 quarters of reliable NAV history. Since mfapi.in returns consistent data from January 2018 and AMFI mandatory direct-plan disclosure commenced in April 2018, the earliest quarter with full feature coverage for a sufficient number of funds is 2020Q2 — yielding 2021Q2 as the first quarter with adequate training observations for a walk-forward model.
- AUM data is approximated from NAV record counts due to mfapi.in not exposing actual assets under management. True AUM from AMFI monthly disclosures would strengthen this feature.
- Portfolio turnover and fund manager tenure, two features listed in the original project specification, were not available through mfapi.in and were not incorporated. These features could be collected from AMFI factsheets.
- The fund universe covers equity mutual funds only. Hybrid and debt funds, which also have regular/direct variants, are out of scope.

8.2 Potential Extensions

- Extending the walk-forward back-test beyond 2021Q2 would require sourcing pre-2018 NAV histories and actual AMFI-reported AUM data not available through mfapi.in. Obtaining these from AMFI factsheets would allow validation across the 2018–2020 period, including the COVID-19 shock of March 2020, providing a more severe stress test of the model's regime robustness.
- Incorporating actual AUM data, portfolio turnover, and manager tenure from AMFI monthly factsheets would likely improve both model performance and the interpretability of SHAP findings.
- Training on a longer history with regime-conditional models (separate models for rate-rising versus rate-falling environments) could improve out-of-sample calibration.
- Extending the precision-at-top-decile analysis to multi-quarter horizons would help determine whether model rankings exhibit persistence across quarters.

9. Conclusion

CommissionLens demonstrates that machine learning can meaningfully predict whether an Indian equity mutual fund's commission charge will be justified by the alpha it generates in the following quarter. Across 3,258 fund-quarter observations spanning 269 equity funds from 2020 to 2024, a Random Forest classifier achieved an AUC-ROC of 0.69 on the held-out test set, and when restricted to the model's top-decile recommendations, 29.6 percent of selected funds genuinely justified their commission, representing a 1.46x lift over uninformed selection.

A fully walk-forward SIP simulation over 36 months (April 2021 to March 2024, covering the 2022 correction and 2023 recovery) produced an annualised XIRR of 24.36% for model-guided selection versus 18.61% for naive regular plan investing a gap of 5.75 percentage points with ₹2,51,784 final corpus against ₹2,33,381 on identical ₹1,80,000 of invested capital. Notably, this advantage is wider than the 3.16 percentage point gap observed in the original 14-month holdout window, indicating that the model's selection signal held up through the 2022 bear phase rather than relying solely on bull-market momentum.

SHAP analysis reveals that recent alpha momentum is the single strongest predictor of commission justification, followed by return volatility and SEBI fund category. Notably, the average expense gap ranks fifth rather than first under Random Forest, a finding that reflects the model's ability to look beyond the cost dimension and identify funds where genuine skill compensates for it. The strong negative SHAP correlation between recent_alpha and ret_std (−0.50) confirms that the model rewards consistent, low-volatility outperformance rather than high-beta return chasing. The framework is deployed as a real-time Streamlit dashboard enabling any investor to obtain a commission-justification score and SHAP-based explanation for any Indian equity mutual fund using live NAV data.

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